

RUDRA PATEL

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Education

LDRP Institute of Technology and Research

Bachelor of Engineering in Computer Engineering (CGPA: 7.58)

Oct 2021 – May 2025

Gujarat, India

Skills

Languages & DB: SQL, HTML, CSS, JavaScript, Python, Go, Rust, PostgreSQL, MySQL, MongoDB

Frameworks: Tauri, Django, Flask, Streamlit

Libraries: ReactJS, ChartJS, Plotly, Pandas, NumPy, Scikit-learn

Tools: Git/GitHub, Docker, Postman, Trello, Jupyter Notebook

Work Experience

Full-Stack Developer, Electrosine Technologies

Nov 2025 – Present

- Developed a scalable cloud backend in Go (Fiber) to manage distributed diagnostic data and sync device configs. Designed multi-tenant RESTful APIs for customers, facilities, equipment, and fault-frequency generation. Used SQLC and automated PostgreSQL migrations for reliable schema management and cross-device data consistency.
- Engineered the frontend for a multi-tenant web portal (React, TypeScript, Tailwind) to monitor distributed motor fleets across facilities. Created modular dashboards (Plotly, Recharts) for 6-hour electrical trends, historical sparklines, and 14-day baseline thresholds, plus admin modules for customers, facilities, equipment nameplates, and power-transmission sequences with debounce logic and role-based access control.
- Architected a cross-platform desktop app (Tauri, React, TypeScript) for offline ESA/MCSA analysis. Built interactive Plotly visualizations for real-time 3-phase waveforms, demodulated currents, and FFT spectrums with fault-frequency overlays. Managed sidecar processes for embedded binary communication, hardware config flows, and automated client-side DOCX reporting.
- Maintained an embedded Rust backend for high-frequency telemetry and on-device ESA, calculating fault frequencies for bearings, gearboxes, eccentricity, and rotor bars. Handled local persistence and updates via SQLite migrations and device-side data pools.

Data Science Intern, Prodigy InfoTech – [Link](#)

Sept 2024 – Oct 2024

- Implemented a decision tree classifier (85% accuracy) for marketing data and modeled social media sentiment shifts (82% accuracy). Visualized categorical features in the Titanic dataset to identify a 74% demographic survival rate, and mapped traffic accident hotspots to reveal primary triggers (70% poor roads, 60% night).

Projects

Expense Tracker Web Application using Django – [GitHub](#)

Mar 2025

[Python, Django, PostgreSQL, HTML, CSS, JS]

- Designed and enhanced a full-stack expense management system with user authentication and CRUD functionality. Utilized PostgreSQL as the backend database for efficient data storage and retrieval. Generated monthly and yearly expense summaries using Django ORM and query optimization. Built a responsive front-end with Bootstrap to ensure compatibility across devices.

Hand Gesture Recognition System – [GitHub](#)

Dec 2024

[Python, OpenCV, MediaPipe, Streamlit]

- Implemented a real-time hand gesture recognition system using Python, OpenCV, and MediaPipe, achieving 97% accuracy. Trained custom ML classifiers for static and dynamic gestures, processing 19 3D landmarks at 30+ FPS. Created an interactive web interface using Streamlit for real-time parameter tuning and gesture training. Enabled gesture-based controls, optimizing performance for smooth real-time processing.

House Price Prediction – [GitHub](#)

Aug 2024

[Python, Flask, Scikit-Learn, Pandas, NumPy, Matplotlib, Seaborn]

- Developed a house price prediction system using Python, Flask, and Scikit-Learn, achieving 85%+ accuracy. Benchmarked and evaluated various regression models and fine-tuned hyperparameters to optimize accuracy. Utilized Pandas, NumPy, and Seaborn for data pre-processing, feature engineering, and visualization.

Facial Recognition with Attendance System – [GitHub](#)

Jan 2024

[Python, Tkinter, OpenCV, MySQL]

- Engineered a real-time face recognition system with 90%+ accuracy, leveraging OpenCV and the face recognition library. Used a Tkinter-based GUI for effortless user registration and image storage. Integrated MySQL database for attendance tracking, automatically logging timestamps. Optimized image processing to recognize faces in under 2 seconds.

Certificates

NPTEL Database Management System

Mar 2024

Secured **Elite Category** certification in Database Management Systems through NPTEL Exam.

Python And Django Framework For Beginners Complete Course - Udemy

May 2025

A comprehensive course on Python and Django, covering core programming, OOP, and full-stack web development.

HackerRank - SQL

Apr 2025

Achieved Verified Certification in SQL Basic, Intermediate and Advanced by HackerRank.